Does Employer-Based Immigration Enforcement Reduce the Undocumented Immigrant Population? Re-examining the Effects of LAWA

Samyam Shrestha* Hugo Sant'Anna†

May 7, 2026

Abstract

This paper performs both a narrow and wide replication of Bohn, Lofstrom, and Raphael, 2014 (*Review of Economics and Statistics*, 2014), who use the synthetic control method to estimate the effect of Arizona's 2007 Legal Arizona Workers Act (LAWA) on the state's unauthorized immigrant population. Our narrow replication confirms their main finding: a 1.5–2 percentage point decline in the share of noncitizen Hispanics following LAWA's implementation. In a wide replication, we re-estimate the treatment effect using the augmented synthetic control method and synthetic difference-in-differences, extend the sample through 2015, and identify both margins of the population response using American Community Survey bilateral migration data. The core finding is robust across all three estimators; however, we find that the effect attenuates after 2011, suggesting partial reversion over the longer horizon. A Poisson gravity difference-in-differences further reveals that both outflows from Arizona and inflows to the state responded to the law, with outflows responding at signing and inflows at implementation.

Keywords: LAWA, immigration enforcement, undocumented immigrants, E-Verify, synthetic control

JEL Codes: C23, J15, J61, K37

*Department of Agricultural and Applied Economics, University of Georgia. Email: samyam@uga.edu.

†Collat School of Business, University of Alabama at Birmingham. Email: hsantanna@uab.edu.

1 Introduction

Over the past few decades, interior immigration enforcement in the U.S. has intensified considerably. Enforcement approaches fall into a few broad categories. The first is police-based enforcement, where law enforcement directly targets and apprehends undocumented individuals. The second is government-service-based enforcement, where access to public benefits or services such as driver's licenses, school enrollment, healthcare, and public housing is conditioned on proving legal status. A distinct third approach is labor-market-based enforcement, in which employers are required to verify the work authorization of new hires, most notably through the E-Verify system. Because this approach works indirectly through hiring decisions rather than direct contact with immigrants, it may produce meaningfully different outcomes than the other two strategies.

One of the clearest examples of this labor-market approach in action is the 2007 Legal Arizona Workers Act (LAWA). The law mandated that all employers in Arizona participate in E-Verify and imposed sanctions on firms found to knowingly hire unauthorized workers. By raising the costs of employing undocumented individuals, LAWA was designed to shift labor demand away from unauthorized immigrants through the market itself, rather than through direct enforcement.

This paper replicates and extends Bohn, Lofstrom, and Raphael, 2014, which examines the effects of LAWA on the likely undocumented population in Arizona. Using the Synthetic Control Method (SCM), the paper finds that LAWA led to a significant reduction in the share of Arizona residents most likely to be unauthorized. We begin with a narrow replication, confirming the paper's main findings. While our estimates differ marginally, they are consistent in direction, magnitude, and significance. We also assess the implementation-robustness of the headline synthetic control finding by re-estimating the regression across seven independent SCM software implementations in R and Python, following the approach of Becker and Klößner (2017) in their replication of Pinotti, 2015. The estimated treatment effects are highly consistent across implementations and closely mirror the original results.

We then extend the analysis in several directions. First, we apply recently developed alternatives to SCM, including the augmented synthetic control method (Ben-Michael, Feller, and Rothstein, 2021a) and synthetic difference-in-differences (Arkhangelsky et al., 2021a), both of which offer improved bias correction and more reliable inference than standard SCM. Second, we extend the sample through 2015, beyond the original paper's window of up to 2009, to assess how the effect evolves over a longer horizon. The original finding holds across all alternative estimators, though the extended sample reveals substantial attenuation after 2011.

Third, we decompose the reduction in the share of likely undocumented immigrants into inflow and outflow margins using the American Community Survey (ACS). We use the ACS to construct estimates of bilateral state-level Hispanic immigrant migration flows, then estimate a Poisson Pseudo-Maximum Likelihood gravity difference-in-differences model to quantify how much of the Arizona immigrant population decline reflects redirected interstate migration versus deterred inflow.

The evidence sharpens the substantive contribution of Bohn, Lofstrom, and Raphael, 2014 along three dimensions: the population effect is robust to implementation choices, operates through both outflow displacement and inflow deterrence with distinct timing, and while sustained through 2015, attenuates after 2011, pointing to partial reversion over the longer run.

2 Replication in a Narrow Sense

Bohn, Lofstrom, and Raphael, 2014 investigates the impact of LAWA using the synthetic control method (SCM) developed by Abadie, Diamond, and Hainmueller (2010). Arizona is treated beginning in 2008, and the authors construct a synthetic control as a weighted average of other U.S. states chosen to match Arizona's pre-treatment characteristics and outcome trajectory.

Formally, let Y_{st} denote the outcome for state s in year t. SCM chooses weights $w_j \geq 0$ (summing to one) on donor states j to minimize the pre-treatment discrepancy:

$$\min_{w} \sum_{t \leq T_0} \left(Y_{\mathrm{AZ},t} - \sum_{j \in \mathcal{D}} w_j Y_{jt} \right)^2, \tag{1}$$

where T_0 is the last pre-LAWA year. The estimated treatment effect at time $t > T_0$ is the gap between Arizona and its synthetic control:

$$\widehat{\tau}_t = Y_{\mathrm{AZ},t} - \sum_{j \in \mathcal{D}} w_j Y_{jt}. \tag{2}$$

Statistical significance is assessed using permutation (placebo) tests that iteratively assign treatment to donor states and compare Arizona's post-treatment gap to the placebo distribution.

In this subsection, we provide replication results using the original methods and samples used in the paper. Figure 1 plots Arizona's noncitizen Hispanic share (ages 15–45, high school diploma or less) alongside its synthetic control from 1998 to 2009. The two series track closely throughout the pre-treatment period, diverge sharply after 2007, and remain separated through 2009. The synthetic control assigns dominant weight to three states: New Mexico (0.46), California (0.43), and Nevada (0.11), with all remaining states receiving near-zero weight. The pre-treatment root mean squared prediction error is small, confirming that the synthetic control closely reproduces Arizona's pre-LAWA trajectory.

Table 1 reports the difference-in-differences estimates across several outcome definitions and pre-period windows. For the main outcome—noncitizen Hispanics aged 15–45 with a high school diploma or less—the DD estimate using the full 1998–2006 pre-period is -0.022, with Arizona ranked 1st out of 47 states in the permutation distribution ($p = 0.022$). The result is consistent across broader population definitions: the effect on all noncitizen Hispanics aged 15 and older is -0.017 ($p = 0.023$), and the effect on noncitizen Hispanics in the full resident population is -0.015 ($p = 0.021$). These magnitudes align closely with the estimates reported in Bohn, Lofstrom, and Raphael, 2014.

Figure 1(b) displays the permutation test. Arizona's post-treatment gap is the most negative in the distribution, lying well below the placebo gaps for all 46 donor states. As a falsification exercise, we apply the same procedure to naturalized Hispanic citizens aged 15–45 with low education. The DD estimate for this placebo outcome is 0.001 ($p = 0.745$), confirming that LAWA affected noncitizen populations specifically.

3 Replication in a Wide Sense

3.1 Augmented Synthetic Control Method

To assess the robustness of the baseline SCM estimates, we implement the Augmented Synthetic Control Method (ASCM) proposed by Ben-Michael, Feller, and Rothstein (2021a). ASCM augments the standard synthetic control with a ridge-regularized outcome model estimated on the donor pool, correcting for bias when pre-treatment fit is imperfect and allowing limited extrapolation beyond the convex hull. The ridge penalty is selected via cross-validation. This bias correction directly addresses the concern that standard SCM estimates may be sensitive to the convex-hull restriction or finite-sample imbalance.

Table 2, Panels D–F, reports the ASCM estimates. For the main outcome—noncitizen Hispanics aged 15–45 with a high school diploma or less—the ASCM DD estimate using the full pre-period is -0.021 ($p = 0.021$), virtually identical to the SCM estimate of -0.022. The broader noncitizen Hispanic population aged 15–45 shows a DD of -0.027 ($p = 0.021$), and the all-ages noncitizen Hispanic share declines by 0.016 ($p = 0.021$). The ridge-based bias correction shifts the estimates by at most 0.001 in absolute value relative to SCM, indicating that the original synthetic control achieved adequate pre-treatment fit and that extrapolation beyond the convex hull of the donor pool is minimal. The ASCM results reinforce the SCM findings: the magnitude of the estimated LAWA effect is not an artifact of the convex-hull restriction or finite-sample bias in the standard synthetic control.

3.2 Synthetic Difference-in-Differences

As an additional robustness exercise, we implement the synthetic difference-in-differences (SDID) estimator of Arkhangelsky et al. (2021a). SDID combines the reweighting logic of synthetic control with the double-differencing structure of DID. The estimator constructs nonnegative unit weights and time weights that balance pre-treatment outcomes, then estimates the treatment effect via a weighted two-way fixed effects regression. This hybrid design makes SDID more tolerant of imperfect pre-treatment fit and latent factor structures than either method alone.

Table 2, Panels G–I, reports the SDID estimates. For the main outcome, the SDID DD using the full pre-period is -0.027 ($p = 0.021$), and the broader noncitizen Hispanic 15–45 group shows -0.030 ($p = 0.021$). All outcomes produce negative estimates with Arizona ranked 1st in the permutation distribution. The SDID point estimates are slightly larger in magnitude than SCM and ASCM because SDID's unit and time weights produce a different counterfactual that more aggressively down-weights pre-treatment periods where Arizona's gap was already slightly negative (see section C.3).

3.3 Extended Time Period

Bohn, Lofstrom, and Raphael (2014) end their analysis in 2009, largely because SB1070 was enacted in 2010, making it difficult to separately distinguish the effects of LAWA from those of the subsequent omnibus immigration policy. However, Amuedo-Dorantes and Lozano (2015) find that SB1070 had minimal independent effects on the noncitizen Hispanic population in Arizona. We therefore extend the sample through 2015, adding six years of post-treatment data to assess the longer-run persistence of LAWA's effects.

The right half of Table 2 reports the extended (1998–2015) estimates alongside the original period. For the main outcome—noncitizen Hispanics aged 15–45 with a high school diploma or less—the SCM DD estimate moves from -0.022 to -0.020, a negligible change. The ASCM estimate is similarly stable at -0.021 in both periods. The SDID estimate, by contrast, grows substantially in magnitude, from -0.027 to -0.038, as additional post-treatment periods receive weight.

Figure 2 plots Arizona against its synthetic control over the full 1998–2015 window. The two series diverge sharply after 2007, with the gap reaching its trough around 2009–2011 (approximately -0.03). After 2011, however, the gap narrows steadily, falling to roughly -0.005 by 2015—within the range of pre-treatment variation. The pattern suggests partial reversion of LAWA's initial effect over the longer horizon.

The attenuation after 2011 raises the question of whether LAWA's impact was a level shift that partially unwound as enforcement salience faded, or whether the closing gap reflects compositional changes in the CPS sample or differential trends unrelated to LAWA. The average post-treatment effect remains negative across all three estimators, but the year-by-year trajectory in Figure A1 indicates that the magnitude diminishes substantially over time.

Because the extended sample overlaps with Arizona's SB 1070 law (key provisions effective 2011), a natural question is whether the post-2011 attenuation reflects the fading of LAWA's effect or the confounding arrival of SB 1070. To disentangle the two, we estimate treatment effects separately for the LAWA-only period (2008–2010, before SB 1070) and the post-SB 1070 period (2011–2015; see table A5). All three estimators agree that the decline is concentrated in the immediate post-LAWA window (estimates from -0.015 to -0.030), while the post-SB 1070 marginal effects are small, mixed in sign, and statistically insignificant. The initial effect is thus attributable to LAWA, not SB 1070, and the post-2011 attenuation is consistent with partial reversion of LAWA's impact rather than an offsetting SB 1070 effect.

3.4 Mechanism Decomposition: Outflow and Inflow

The CPS-based synthetic control evidence pins down the magnitude of the population effect but cannot reveal whether the decline reflects existing residents leaving Arizona or prospective migrants choosing alternative destinations. Distinguishing the two channels matters for interpretation: outflow displacement is geographic redistribution within the unauthorized population, whereas inflow deterrence operates ex ante and would persist even at reduced enforcement intensity. We use American Community Survey microdata 2005–2010 to estimate both margins directly. Each respondent's state of residence one year prior (`migplac1`) is collapsed by origin, destination, and ACS year into a population-weighted bilateral flow flow_{ijt} for working-age Hispanic noncitizens with high-school diploma or less—the same demographic group as the main analysis. We estimate a gravity Poisson Pseudo-Maximum Likelihood difference-in-differences (Santos Silva and Tenreyro, 2006) with origin-destination pair, origin-by-year, and destination-by-year fixed effects:

$$\text{flow}_{ijt} = \exp\Big(\beta\, T_{ijt} + \alpha_{ij} + \gamma_{i,t} + \delta_{j,t}\Big) \cdot \varepsilon_{ijt}. \tag{3}$$

For outflow, $T_{ijt} = \mathbf{1}\{i = \text{AZ}\} \cdot \mathbf{1}\{j \in \mathcal{N}\} \cdot \mathbf{1}\{t \geq t^*\}$ with $\mathcal{N} = \{\text{CA, NV, NM, UT}\}$; for inflow, $T_{ijt} = \mathbf{1}\{j = \text{AZ}\} \cdot \mathbf{1}\{t \geq t^*\}$. The treatment indicator is identified as a triple interaction even under the saturated gravity FE because lower-order terms are perfectly collinear with the fixed effects. Standard errors are clustered at the pair level. Section B reports specifications, full results, and event studies.

Three findings emerge. First, AZ-to-neighbor flows rose by approximately 94 % post-LAWA-signing under the headline gravity specification ($\hat{\beta} = 0.661$, $p = 0.010$, robust to less saturated fixed effects and to log-OLS functional-form). Pre-trend Wald tests pass under both pair-and-year and full gravity FE. Second, inflows to Arizona from any origin fell by 36 % ($\hat{\beta} = -0.448$, $p = 0.061$) under the implementation-only timing, and the symmetric mirror specification (neighbor-state $\to$ AZ) yields -58 % ($p < 0.001$). Third, the dynamic patterns differ informatively: outflow jumps in the LAWA-signing window (ACS 2008, capturing calendar-2007 flows after the July 2007 signing), whereas inflow drops one year later when enforcement begins (ACS 2009 onward). This timing asymmetry is consistent with existing residents responding to the announcement during the six-month gap between signing and implementation, while prospective migrants update only when enforcement actually starts.

3.5 Cross-Implementation Robustness

Becker and Klößner (2017) document that the same synthetic control specification can yield economically different estimates when run through different software, because the original Abadie, Diamond, and Hainmueller (2011) optimizer solves a non-convex outer problem in the predictor weight matrix. We re-estimate the headline outcome (proportion of noncitizen Hispanics aged 15–45 with high-school diploma or less) through seven independent implementations on identical inputs: `Synth`, `tidysynth`, `scpi` (Cattaneo, Feng, and Titiunik, 2021), `synthdid`, and `gsynth` in R, plus `pysyncon`'s `Synth` and ridge-augmented `AugSynth` classes in Python. Section D reports the full comparison.

All seven implementations return a negative effect; the five direct synthetic control estimates concentrate in the narrow interval $[-0.022, -0.017]$. The residual variation is interpretable rather than random: implementations that solve only the simplex-constrained problem on outcome lags (`tidysynth`, `scpi`, and `Synth` restricted to outcome lags only) return identical donor weights and an ATE of -0.020, whereas the original `Synth` specification with the BLR predictor list returns -0.022. The within-`Synth` specification choice (covariates versus outcome lags only) accounts for the entire cross-package gap, sharpening Kaul et al. (2022): predictor selection, not solver choice or programming language, drives implementation-level variation in this application. The Cattaneo–Feng–Titiunik prediction interval delivered by `scpi` is $[-0.052, -0.007]$, bounded away from zero on the negative side. The Bohn, Lofstrom, and Raphael (2014) headline finding survives substitution of solver, language, and software team.

4 Conclusion

We replicate the main findings of Bohn, Lofstrom, and Raphael, 2014 and confirm that the 2007 Legal Arizona Workers Act reduced Arizona's noncitizen Hispanic population share by approximately 1.5–2 percentage points. The result holds across three distinct estimators—the original synthetic control method, the augmented synthetic control of Ben-Michael, Feller, and Rothstein (2021a), and the synthetic difference-in-differences of Arkhangelsky et al. (2021a)—and across seven independent SCM software implementations in R and Python. Extending the sample through 2015, however, reveals that while the effect remains negative on average, it attenuates substantially after 2011, with the gap narrowing from roughly 3 percentage points to less than 1 by 2015. As a falsification check, none of the estimators detect effects on naturalized Hispanic citizens.

Beyond replicating the magnitude, we identify the channels through which the effect operates. Using ACS bilateral migration data, a Poisson gravity difference-in-differences estimates a 94 % post-LAWA increase in outflows from Arizona to neighboring states and a 36–58 % decline in inflows to Arizona from any origin. Outflow responds at LAWA's signing in mid-2007; inflow responds one year later when enforcement begins. LAWA thus operated through both margins of the migration response, with the outflow component representing geographic redistribution within the unauthorized population and the inflow component representing ex-ante deterrence of prospective migrants.

Two limitations merit acknowledgment. First, noncitizen Hispanic status is a proxy for unauthorized status. Second, our analysis is specific to Arizona, a border state with a large pre-existing unauthorized population; external validity to states with different labor markets remains an open question. The convergence of three estimators across multiple outcomes, the implementation-robustness across seven SCM software pipelines, and the dual-margin mechanism evidence together strengthen confidence that Bohn, Lofstrom, and Raphael's original finding reflects a genuine causal effect of employer-based immigration enforcement.

At the same time, the attenuation visible in the extended sample suggests that the population response may be partially temporary, a finding that qualifies the original paper's policy implications.

References

Abadie, A. (2021). "Using Synthetic Controls: Feasibility, Data Requirements, and Methodological Aspects". In: *Journal of Economic Literature* 59.2, pp. 391–425. DOI: 10.1257/jel.20191450.

Abadie, A., A. Diamond, and J. Hainmueller (2010). "Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program". In: *Journal of the American Statistical Association* 105.490, pp. 493–505.

Abadie, A., A. Diamond, and J. Hainmueller (2011). "Synth: An R Package for Synthetic Control Methods in Comparative Case Studies". In: *Journal of Statistical Software* 42.13, pp. 1–17. DOI: 10.18637/jss.v042.i13.

Amuedo-Dorantes, C. and F. Lozano (2015). "On the Effectiveness of SB1070 in Arizona". In: *Economic Inquiry* 53.1, pp. 335–351.

Arkhangelsky, D., S. Athey, D. A. Hirshberg, G. W. Imbens, and S. Wager (2021a). "Synthetic Difference-in-Differences". In: *American Economic Review* 111.12, pp. 4088–4118.

Arkhangelsky, D., S. Athey, D. A. Hirshberg, G. W. Imbens, and S. Wager (2021b). "Synthetic Difference-in-Differences". In: *American Economic Review* 111.12, pp. 4088–4118. DOI: 10.1257/aer.20190159.

Becker, M. and S. Klößner (2017). "Estimating the Economic Costs of Organized Crime by Synthetic Control Methods". In: *Journal of Applied Econometrics* 32.7, pp. 1367–1369. DOI: 10.1002/jae.2572.

Ben-Michael, E., A. Feller, and J. Rothstein (2021a). "The Augmented Synthetic Control Method". In: *Journal of the American Statistical Association* 116.536, pp. 1789–1803.

Ben-Michael, E., A. Feller, and J. Rothstein (2021b). "The Augmented Synthetic Control Method". In: *Journal of the American Statistical Association* 116.536, pp. 1789–1803. DOI: 10.1080/01621459.2021.1929245.

Bohn, S., M. Lofstrom, and S. Raphael (2014). "Did the 2007 Legal Arizona Workers Act Reduce the State's Unauthorized Immigrant Population?" In: *Review of Economics and Statistics* 96.2, pp. 258–269.

Cattaneo, M. D., Y. Feng, and R. Titiunik (2021). "Prediction Intervals for Synthetic Control Methods". In: *Journal of the American Statistical Association* 116.536, pp. 1865–1880. DOI: 10.1080/01621459.2021.1979561.

Ferman, B. and C. Pinto (2021). "Synthetic Controls with Imperfect Pretreatment Fit". In: *Quantitative Economics* 12.4, pp. 1197–1221. DOI: 10.3982/QE1596.

Ferman, B., C. Pinto, and V. Possebom (2020). "Cherry Picking with Synthetic Controls". In: *Journal of Policy Analysis and Management* 39.2, pp. 510–532. DOI: 10.1002/pam.22206.

Hoekstra, M. and S. Orozco-Aleman (2017). "Illegal Immigration, State Law, and Deterrence". In: *American Economic Journal: Economic Policy* 9.2, pp. 228–252.

Kaul, A., S. Klößner, G. Pfeifer, and M. Schieler (2022). "Standard Synthetic Control Methods: The Case of Using All Preintervention Outcomes Together with Covariates". In: *Journal of Business & Economic Statistics* 40.3, pp. 1362–1376. DOI: 10.1080/07350015.2021.1930012.

Klößner, S., A. Kaul, G. Pfeifer, and M. Schieler (2018). "Comparative Politics and the Synthetic Control Method Revisited: A Note on Abadie et al. (2015)". In: *Swiss Journal of Economics and Statistics* 154.1, p. 11. DOI: 10.1186/s41937-017-0004-9.

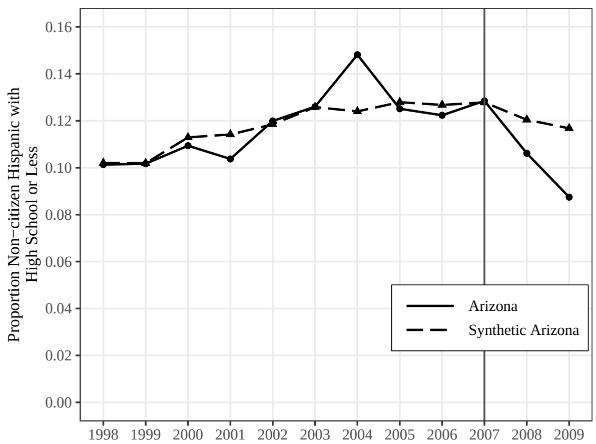
Pinotti, P. (2015). "The economic costs of organised crime: Evidence from Southern Italy". In: *The Economic Journal* 125.586, F203–F232.

Powell, D. (2026). "Imperfect Synthetic Controls". In: *Journal of Applied Econometrics* 41.3, pp. 253–264. DOI: 10.1002/jae.70035.

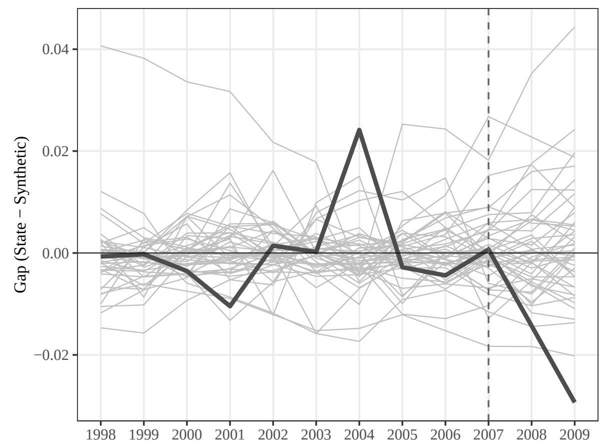
Santos Silva, J. M. C. and S. Tenreyro (2006). "The Log of Gravity". In: *Review of Economics and Statistics* 88.4, pp. 641–658. DOI: 10.1162/rest.88.4.641.

Xu, Y. (2017). "Generalized Synthetic Control Method: Causal Inference with Interactive Fixed Effects Models". In: *Political Analysis* 25.1, pp. 57–76. DOI: 10.1017/pan.2016.2.

Figures and Tables



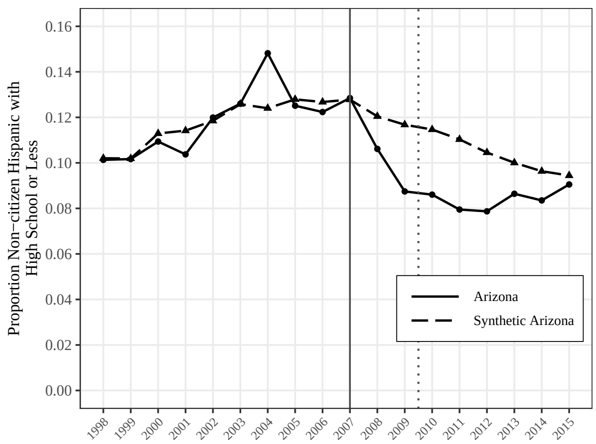
(a) Arizona vs. Synthetic Arizona, 1998–2009



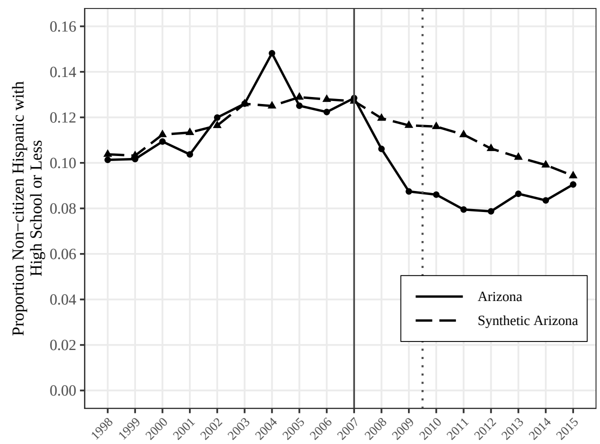
(b) Permutation test: Arizona (thick line) vs. 46 placebos

Figure 1: Synthetic control estimates of the effect of LAWA on the proportion of noncitizen Hispanics with a high school diploma or less among prime working-age persons, 1998–2009

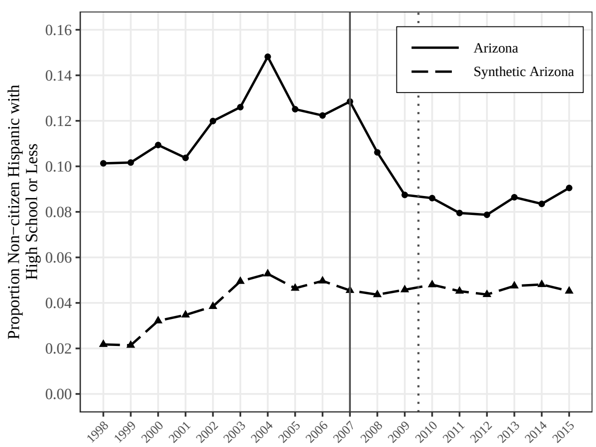
Notes: Panel (a) plots the outcome for Arizona and its synthetic control constructed from the donor pool of 46 states, excluding states with broad E-Verify mandates (MS, RI, SC, UT). Panel (b) plots the gap (state minus synthetic) for all states; Arizona is shown with the thick black line. The vertical dashed line marks 2007, the year LAWA was signed.



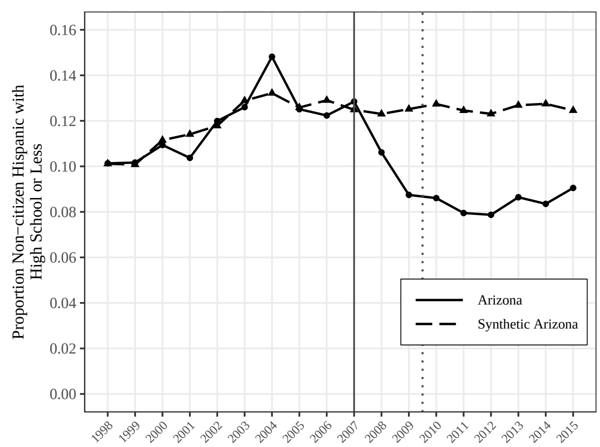
(a) SCM



(b) Augmented SCM



(c) Synthetic DiD



(d) Synthetic DiD (demeaned)

Figure 2: Arizona vs. Synthetic Arizona over the extended period, 1998–2015

Notes: Panels (a)–(c) plot Arizona and its synthetic control for the main outcome (proportion of noncitizen Hispanics aged 15–45 with a high school diploma or less) over the extended 1998–2015 window under three estimators: (a) the original SCM of Abadie, Diamond, and Hainmueller, 2010, (b) the augmented SCM of Ben-Michael, Feller, and Rothstein (2021a), and (c) the synthetic difference-in-differences of Arkhangelsky et al. (2021a). Panel (d) replots the SDID counterfactual after adding the mean pre-treatment gap to the synthetic line, so that pre-treatment levels align and the post-treatment divergence is visible on the same scale as panels (a)–(b). The solid vertical line marks 2007 (LAWA signed); the dotted vertical line marks 2009, the endpoint of the original Bohn, Lofstrom, and Raphael, 2014 analysis.

Table 1: Narrow Replication: Difference-in-Differences Estimates of the Effect of LAWA on Population Shares, 1998–2009

	Mean difference: AZ − Synthetic AZ			DD using pre 1998–2006			DD using pre 2005–2006		
Outcome	1998–2006	2005–2006	2008–2009	DD	Rank	p	DD	Rank	p
Panel A: Prime working-age population (15–45)									
Noncitizen Hispanic	0.000	−0.005	−0.027	−0.027	1/47	0.022	−0.022	2/47	0.043
Noncitizen Hispanic, HS or less	0.000	−0.004	−0.022	−0.022	1/47	0.022	−0.018	1/47	0.022
Panel B: Population age 15 and over									
Noncitizen Hispanic	0.000	−0.001	−0.017	−0.017	1/47	0.023	−0.016	1/47	0.023
Noncitizen Hispanic, HS or less	−0.001	0.000	−0.013	−0.013	1/47	0.022	−0.013	1/47	0.022
Panel C: Entire resident population									
Noncitizen Hispanic	0.000	−0.002	−0.015	−0.015	1/47	0.021	−0.013	1/47	0.021

Notes: Average differences between Arizona and its synthetic control in the pre- and post-treatment periods. DD denotes the difference-in-differences estimate. The one-tailed p-value uses the empirical distribution of placebo DD estimates from 46 donor states. States with broad E-Verify mandates (MS, RI, SC, UT) are excluded from the donor pool. Data: Monthly CPS, 1998–2009.

Table 2: Comprehensive Difference-in-Differences Estimates: SCM, SDID, and ASCM across Population Groups and Time Periods

	Mean difference: AZ − Synth. AZ				Original period (post: 2008–2009)						Extended period (post: 2008–2015)					
					DD (pre 1998–2006)			DD (pre 2005–2006)			DD (pre 1998–2006)			DD (pre 2005–2006)		
Outcome	1998–2006	2005–2006	2008–2009	2008–2015	DD	Rank	p	DD	Rank	p	DD	Rank	p	DD	Rank	p
Synthetic Control Method (SCM)																
Panel A: Prime working-age (15–45)																
Noncitizen Hispanic	0.000	−0.005	−0.027	−0.026	−0.027	1/46	0.022	−0.022	2/46	0.043	−0.027	1/46	0.022	−0.022	1/46	0.022
Noncitizen Hispanic, HS or less	0.000	−0.004	−0.022	−0.020	−0.022	1/46	0.022	−0.018	1/46	0.022	−0.020	1/46	0.022	−0.016	1/46	0.022
Panel B: Population 15 and over																
Noncitizen Hispanic	0.000	−0.001	−0.017	−0.015	−0.017	1/45	0.022	−0.017	1/45	0.022	−0.015	1/45	0.022	−0.014	1/45	0.022
Noncitizen Hispanic, HS or less	−0.001	0.000	−0.013	−0.011	−0.013	1/45	0.022	−0.013	1/45	0.022	−0.010	1/45	0.022	−0.010	1/45	0.022
Panel C: All residents																
Noncitizen Hispanic	0.000	−0.002	−0.015	−0.014	−0.015	1/47	0.021	−0.013	1/47	0.021	−0.013	1/47	0.021	−0.012	1/47	0.021
Augmented Synthetic Control (ASCM)																
Panel D: Prime working-age (15–45)																
Noncitizen Hispanic	0.000	−0.005	−0.027	−0.027	−0.027	1/47	0.021	−0.022	2/47	0.043	−0.027	1/47	0.021	−0.022	1/47	0.021
Noncitizen Hispanic, HS or less	0.000	−0.005	−0.021	−0.021	−0.021	1/47	0.021	−0.017	1/47	0.021	−0.021	1/47	0.021	−0.016	1/47	0.021
Panel E: Population 15 and over																
Noncitizen Hispanic	0.000	−0.002	−0.015	−0.012	−0.015	1/47	0.021	−0.013	1/47	0.021	−0.012	1/47	0.021	−0.010	1/47	0.021
Noncitizen Hispanic, HS or less	0.000	−0.001	−0.012	−0.009	−0.012	1/47	0.021	−0.011	1/47	0.021	−0.009	1/47	0.021	−0.008	2/47	0.043
Panel F: All residents																
Noncitizen Hispanic	0.000	−0.001	−0.016	−0.012	−0.016	1/47	0.021	−0.015	1/47	0.021	−0.012	1/47	0.021	−0.011	1/47	0.021
Synthetic Difference-in-Differences (SDID)																
Panel G: Prime working-age (15–45)																
Noncitizen Hispanic	0.000	−0.004	−0.031	−0.042	−0.030	1/47	0.021	−0.026	1/47	0.021	−0.042	1/47	0.021	−0.038	1/47	0.021
Noncitizen Hispanic, HS or less	0.000	−0.003	−0.027	−0.038	−0.027	1/47	0.021	−0.023	1/47	0.021	−0.038	1/47	0.021	−0.034	1/47	0.021
Panel H: Population 15 and over																
Noncitizen Hispanic	0.000	−0.001	−0.020	−0.024	−0.019	1/47	0.021	−0.018	1/47	0.021	−0.024	1/47	0.021	−0.023	1/47	0.021
Noncitizen Hispanic, HS or less	0.000	−0.001	−0.017	−0.021	−0.017	1/47	0.021	−0.016	1/47	0.021	−0.021	1/47	0.021	−0.021	1/47	0.021
Panel I: All residents																
Noncitizen Hispanic	0.000	−0.001	−0.019	−0.024	−0.019	1/47	0.021	−0.018	1/47	0.021	−0.024	1/47	0.021	−0.023	1/47	0.021

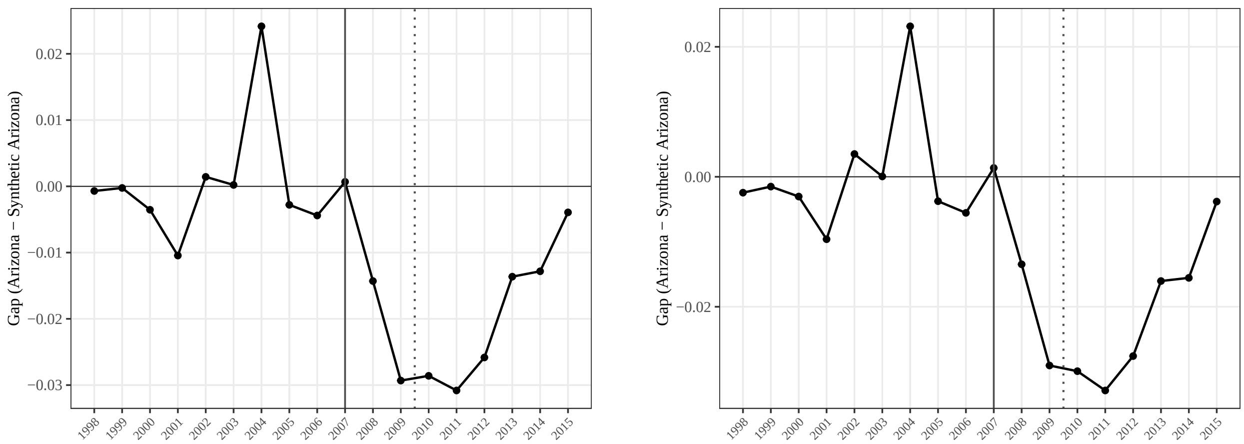
Notes: Each cell reports the difference-in-differences estimate computed as the average post-treatment gap (Arizona minus its counterfactual) minus the average pre-treatment gap. Panels A–C use the standard synthetic control method (Abadie, Diamond, and Hainmueller, 2010); Panels D–F use the augmented synthetic control (Ben-Michael, Feller, and Rothstein, 2021a); Panels G–I use synthetic difference-in-differences (Arkhangelsky et al., 2021a). The counterfactual gap is constructed from each method's fitted weights. Rank and p-values are from permutation (placebo) tests: each donor state is iteratively assigned as treated, the same DD is computed, and Arizona is ranked (most negative = rank 1). One-tailed $p = \text{rank}/N$. States with broad E-Verify mandates (MS, RI, SC, UT) are excluded from the donor pool. 2007 is excluded as a transition year. Data: Monthly CPS, 1998–2015.

A Additional Figures and Tables

Table A1: Trends in the Proportion of Arizona Residents Who Are Hispanic Noncitizens, by Age and Education, 1998–2015

	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
Population proportions																		
Hisp. noncitizen (all ages)	0.082	0.083	0.085	0.084	0.085	0.085	0.100	0.089	0.092	0.093	0.078	0.066	0.065	0.061	0.061	0.065	0.068	0.077
Hisp. noncitizen, 15+	0.093	0.093	0.096	0.093	0.099	0.099	0.115	0.104	0.106	0.109	0.092	0.080	0.079	0.074	0.075	0.078	0.083	0.094
By education, 15+																		
Less than high school	0.067	0.069	0.065	0.060	0.062	0.062	0.076	0.064	0.064	0.069	0.060	0.047	0.045	0.044	0.049	0.050	0.053	0.058
High school graduate	0.013	0.013	0.019	0.017	0.021	0.024	0.023	0.027	0.028	0.026	0.019	0.020	0.022	0.019	0.015	0.018	0.019	0.024
Some college or more	0.013	0.011	0.012	0.016	0.016	0.014	0.016	0.013	0.014	0.014	0.013	0.012	0.013	0.011	0.010	0.010	0.011	0.012
By education, 15–45																		
Hisp. noncitizen, 15–45	0.121	0.115	0.126	0.125	0.140	0.146	0.171	0.144	0.143	0.148	0.124	0.104	0.102	0.096	0.093	0.100	0.098	0.107
Less than high school	0.084	0.085	0.083	0.079	0.089	0.090	0.111	0.086	0.080	0.089	0.078	0.062	0.054	0.053	0.056	0.061	0.058	0.056
High school graduate	0.018	0.017	0.027	0.024	0.031	0.036	0.037	0.039	0.042	0.039	0.028	0.025	0.032	0.026	0.023	0.026	0.026	0.034
Some college or more	0.019	0.014	0.016	0.022	0.021	0.020	0.023	0.019	0.020	0.020	0.018	0.017	0.016	0.016	0.014	0.014	0.015	0.016

Notes: Tabulated using all monthly Current Population Survey files between 1998 and 2015. Each cell reports the proportion of Arizona's population in the indicated demographic group. Education categories sum to the group total (minor rounding discrepancies possible). Values for 2003–2009 match Bohn, Lofstrom, and Raphael, 2014 Table 1 exactly; values for 2000–2002 differ by 0.005–0.007 due to post-2010-census population weight revisions in later IPUMS CPS extracts. The key patterns—rising shares through 2006, sharp decline in 2008–2009—are identical across extracts.



(a) SCM

(b) Augmented SCM

Figure A1: Gap between Arizona and Synthetic Arizona, 1998–2015

Notes: Each panel plots the difference between Arizona's outcome and its synthetic control (Arizona minus Synthetic) over the extended 1998–2015 period under (a) SCM and (b) ASCM. SDID is omitted because its unit weights do not produce a level-matched counterfactual; the SDID treatment effect is identified through a DiD intercept adjustment rather than the level gap. The solid vertical line marks 2007 (LAWA signed); the dotted vertical line marks 2009, the endpoint of the original Bohn, Lofstrom, and Raphael, 2014 analysis. Outcome: proportion of noncitizen Hispanics with a high school diploma or less among persons aged 15–45.

Table A2: Falsification Test: Naturalized Hispanic Citizens

	1998–2009			1998–2015		
Outcome	SCM	ASCM	SDID	SCM	ASCM	SDID
All ages	0.0006	0.0009	0.0010	0.0027	0.0028	0.0044*
	[0.766]	[0.730]	(0.0019)	[0.213]	[0.521]	(0.0024)
Age 15–45	0.0000	−0.0003	−0.0016	0.0017	−0.0001	0.0023
	[1.000]	[0.994]	(0.0017)	[0.340]	[0.552]	(0.0027)
Age 15–45, HS or less	0.0011	0.0009	−0.0012	0.0021	0.0025	0.0010
	[0.553]	[0.459]	(0.0013)	[0.213]	[0.268]	(0.0017)

Notes: Estimates comparing Arizona to its synthetic control for naturalized Hispanic citizens, a population not subject to employer verification mandates. SCM reports the difference-in-differences estimate (no standard inference available). ASCM reports the average ATT from augmented synthetic control with ridge bias correction; conformal p-values in brackets. SDID reports the ATT from synthetic difference-in-differences; standard errors in parentheses computed via the placebo method. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The absence of significant effects across all three estimators and both time windows confirms that the main results are specific to noncitizen populations. U.S.-born Hispanics are omitted because their population share mechanically increases when noncitizen Hispanics leave the state, making them unsuitable as a falsification group.

B Decomposing the Population Effect: Outflow and Inflow

The synthetic control evidence confirms that LAWA reduced Arizona's noncitizen Hispanic population share, but the CPS-based analysis cannot reveal whether the decline came from outflow displacement (Arizona residents leaving) or inflow deterrence (prospective migrants choosing other destinations). The two channels carry different policy implications: outflow displacement is geographic redistribution within the unauthorized population, whereas inflow deterrence operates ex ante and would persist even at reduced enforcement intensity. This section deploys ACS bilateral migration data to estimate both channels directly through a single gravity-DiD framework on the same subjects (working-age Hispanic noncitizens with high-school diploma or less, the same group as in the main analysis).

B.1 ACS Bilateral Flows: Specification

We use ACS microdata 2005–2010 with the `migplac1` variable, which records each respondent's state of residence one year before the survey. Restricting the sample to working-age (16–64) Hispanic noncitizens with a high-school diploma or less, we collapse to bilateral migration flows by origin state, destination state, and ACS year, with each respondent contributing their ACS person weight (`perwt`) to the flow count. The dependent variable flow_{ijt} is a population-representative count of intra-period migrants from origin i to destination j in ACS year t. Because ACS year t measures migration during calendar year $t-1$, ACS 2008 captures flows during 2007 (the LAWA-signing year, July 2, 2007), ACS 2009 captures the first full implementation year, and ACS 2010 captures the second.

For both outflow and inflow we estimate gravity DiD models by Poisson Pseudo-Maximum Likelihood (PPML; Santos Silva and Tenreyro, 2006) with standard errors clustered at the origin-destination pair level. The general specification is

$$\text{flow}_{ijt} = \exp(\beta\, T_{ijt} + \alpha_{ij} + \gamma_{i,t} + \delta_{j,t}) \cdot \varepsilon_{ijt}, \tag{4}$$

where α_{ij} is a pair fixed effect absorbing time-invariant gravity (distance, language, network ties), $\gamma_{i,t}$ is an origin-by-year fixed effect absorbing origin-state-year shocks, and $\delta_{j,t}$ is a destination-by-year fixed effect absorbing destination-state-year shocks. Under this fixed-effects structure, the only treatment indicators that survive collinearity are triple interactions of the form (treated origin) $\times$ (treated destination) $\times$ (post). We balance the panel by reconstructing the $51 \times 50 \times 6 = 15{,}300$ origin-destination-year grid (interstate moves only) and filling unobserved cells with zero flow, since PPML's log link handles zeros without truncation.

Outflow specification. We drop rows where destination = AZ (so the regression isolates flows from AZ rather than mixing in inflow), and define $T^{\text{out}}_{ijt} = \mathbf{1}\{i = \text{AZ}\} \cdot \mathbf{1}\{j \in \mathcal{N}\} \cdot \mathbf{1}\{t \geq t^*\}$, with $\mathcal{N} = \{\text{CA, NV, NM, UT}\}$ the contiguous neighbors. The coefficient β^{out} identifies the differential post-LAWA change in AZ$\to$neighbor flows beyond what AZ-origin year shocks and neighbor-destination year shocks would predict.

Inflow specification. We drop rows where origin = AZ. The headline indicator is $T^{\text{in}}_{ijt} = \mathbf{1}\{j = \text{AZ}\} \cdot \mathbf{1}\{t \geq t^*\}$, identified under pair and origin-by-year fixed effects (destination-by-year is dropped because the AZ-destination cells are exactly the treatment cells). The coefficient β^{in} identifies the change in flows toward Arizona from any origin, net of pair gravity and origin-state-year shocks. As a symmetric mirror of the outflow specification, we also report $T_{ijt}^{\text{in,N}} = \mathbf{1}\{i \in \mathcal{N}\} \cdot \mathbf{1}\{j = \text{AZ}\} \cdot \mathbf{1}\{t \geq t^*\}$ with the full gravity FE.

We report two timing cuts: $t^* = 2008$ treats the ACS year capturing 2007 calendar flows as post-treatment, picking up the announcement-period response to LAWA's July 2007 signing; $t^* = 2009$ restricts post-treatment to fully implemented years.

B.2 Results: Outflow

Table A3 reports outflow estimates. The headline gravity PPML with $t^* = 2008$ returns $\hat{\beta}^{\text{out}} = 0.661$ (SE $= 0.255$, $p = 0.010$), implying AZ→neighbor flows were 93.7 % higher post-LAWA than the gravity-FE counterfactual would predict. The estimate is stable across a less-saturated pair-and-year fixed-effects specification ($\hat{\beta} = 0.629$, $p < 0.001$) and a log(1 + flow) OLS functional-form check ($\hat{\beta} = 0.778$, $p = 0.060$). The conservative implementation-only cut ($t^* = 2009$) yields $\hat{\beta} = 0.316$ ($p = 0.39$): once origin-by-year fixed effects absorb the AZ-wide year shocks, the substitution-to-neighbors signal in 2009–2010 alone is too weak to identify precisely. The headline thus depends on including the LAWA-signing-period response, consistent with anticipatory migration after the July 2007 signing.

Figure A2(a) traces dynamics. Pre-treatment 2005 and 2006 coefficients are flat near zero (joint Wald $p = 0.74$ under pair-and-year FE, $p = 0.44$ under full gravity FE). The 2008 ACS coefficient jumps to +0.65 ($p = 0.001$), 2009 attenuates to +0.27, and 2010 jumps again to +0.70 ($p = 0.030$). The post-period coefficients are positive in all three years; outflow responds at signing rather than waiting for implementation.

B.3 Results: Inflow

Table A4 reports inflow estimates. The broad PPML inflow indicator with $t^* = 2008$ returns $\hat{\beta}^{\text{in}} = -0.169$ (SE $= 0.180$, $p = 0.348$): including the announcement-period year (when inflow had not yet adjusted) attenuates the estimate. The implementation-only cut $t^* = 2009$ yields $\hat{\beta}^{\text{in}} = -0.448$ (SE $= 0.239$, $p = 0.061$, -36.1 % inflow), marginally significant at conventional levels. The mirror specification restricting treated origins to AZ's contiguous neighbors returns $\hat{\beta}^{\text{in,N}} = -0.862$ (SE $= 0.252$, $p < 0.001$, -57.8 % inflow from neighbors), suggesting that the inflow channel is concentrated in the same migration corridors that absorbed the outflow.

Figure A2(b) traces inflow dynamics. Pre-treatment 2005 and 2006 coefficients lie near zero (joint Wald $p = 0.72$). The 2008 ACS coefficient remains near zero; the inflow channel does not respond at LAWA's signing. The 2009 coefficient drops sharply to -0.57 ($p = 0.021$), and 2010 stays low at -0.43. The dynamic pattern is informative about behavioral mechanism: existing AZ residents have time to relocate during the announcement window between July 2007 and January 2008, so outflow responds at signing; prospective migrants from elsewhere update once enforcement actually begins, so inflow responds at implementation.

B.4 Interpretation

LAWA operated through both channels of the migration margin. AZ→neighbor outflows rose by approximately 94 % net of common origin and destination shocks, indicating substantial geographic substitution toward neighboring states. Inflows to Arizona fell by 36 % to 58 % depending on specification, indicating real ex-ante deterrence among prospective migrants. The two responses follow different timing patterns: outflow at signing, inflow at implementation. Both pre-trend tests pass, both effects persist through 2010, and the gravity FE structure rules out time-varying origin or destination shocks as confounders.

The decomposition reveals what the CPS-based main result cannot. Outflow displacement to U.S. neighbors is a real component of the population effect, suggesting that some of LAWA's local impact was geographic redistribution rather than aggregate reduction in unauthorized residence—a familiar lesson from the broader sub-federal enforcement literature (Hoekstra and Orozco-Aleman, 2017; Amuedo-Dorantes and Lozano, 2015). Inflow deterrence is the second component, and the magnitudes ($-36\,\%$ to $-58\,\%$) are economically meaningful even though they are imprecisely estimated under the most demanding fixed-effects specification.

It is important to note what the ACS measurement cannot capture. ACS `migplac1` tracks interstate U.S. migration only; international flows from Mexico are outside the data window. A direct quantification of international inflow deterrence would require survey data on intended-migration patterns of the kind Hoekstra and Orozco-Aleman (2017) use for SB 1070. The decomposition presented here should therefore be read as identifying interstate channels, not international ones.

Table A3: ACS Bilateral Flows: Gravity DiD on Outflow from Arizona

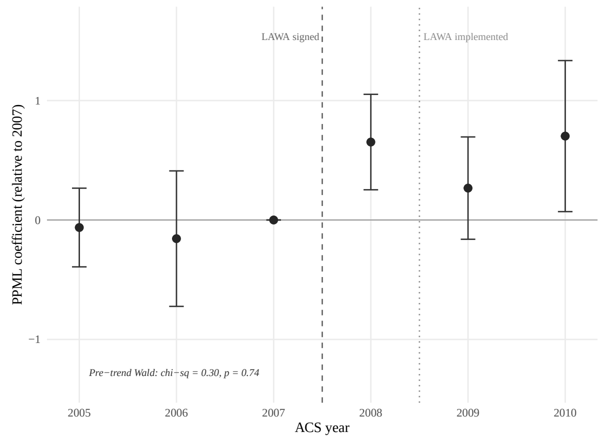
	PPML gravity FE post $\geq$ 2008 (1) headline	PPML gravity FE post $\geq$ 2009 (2) impl. only	log(1+flow) gravity FE post $\geq$ 2008 (3) OLS check	PPML pair+year FE post $\geq$ 2008 (4) less saturated
AZ $\rightarrow$ Neighbors	0.661***	0.316	0.778*	0.629***
	(0.255)	(0.367)	(0.413)	(0.174)
% change in flow	93.7%	37.1%	117.6%	87.5%
Pair FE	Yes	Yes	Yes	Yes
Origin $\times$ year FE	Yes	Yes	Yes	No
Destination $\times$ year FE	Yes	Yes	Yes	No
Year FE	–	–	–	Yes
Cluster (pair)	Yes	Yes	Yes	Yes
Pre-trend Wald p-value (simple FE)			0.740	
Pre-trend Wald p-value (gravity FE)			0.438	
Observations	5,478	5,478	15,000	5,682

Notes: Bilateral migration flows from origin state i to destination state j in ACS year t, restricted to working-age (16–64) Hispanic noncitizens with high-school diploma or less. Inflow rows (destination = AZ) are dropped so the regression isolates outflow from Arizona. Treatment indicator is $T_{ijt}^{\text{out}} = \mathbf{1}\{i = \text{AZ}\} \cdot \mathbf{1}\{j \in \{\text{CA, NV, NM, UT}\}\} \cdot \mathbf{1}\{t \geq t^*\}$. Columns (1)–(3) saturate with pair, origin-by-year, and destination-by-year fixed effects (gravity DiD). Column (4) replaces origin-by-year and destination-by-year with a common year fixed effect (less saturated). PPML estimated by `fixest::fepois`; column (3) is OLS on $\log(1 + \text{flow})$ for functional-form robustness. Standard errors clustered at the pair level in parentheses. Pre-trend Wald p-values are joint tests that the 2005 and 2006 event-study coefficients on AZ$\rightarrow$neighbor pairs equal zero (reference year 2007). Stars: $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

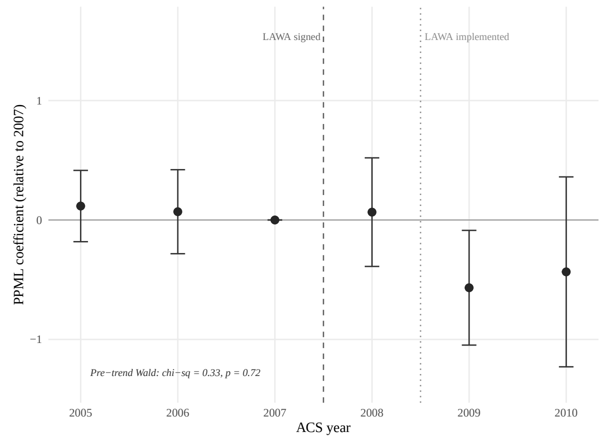
Table A4: ACS Bilateral Flows: PPML on Inflow to Arizona

	PPML broad inflow post $\geq$ 2008 (1) headline	PPML broad inflow post $\geq$ 2009 (2) impl. only	PPML gravity mirror post $\geq$ 2008 (3) neigh. orig.
Inflow to AZ (any origin)	-0.169	-0.448*	—
	(0.180)	(0.239)	
Neighbor origin $\to$ AZ	—	—	-0.862***
			(0.252)
% change in flow	-15.6%	-36.1%	-57.8%
Pair FE	Yes	Yes	Yes
Origin $\times$ year FE	Yes	Yes	Yes
Destination $\times$ year FE	No	No	Yes
Cluster (pair)	Yes	Yes	Yes
Pre-trend Wald p-value (simple FE)		0.716	
Pre-trend Wald p-value (gravity FE)		0.740	
Observations	5,528	5,528	5,437

Notes: Bilateral migration flows for the same demographic group as Table A3, restricted to flows into Arizona by dropping rows where origin = AZ. Columns (1)–(2) report the broad inflow specification: $T^{\text{in}}_{ijt} = \mathbf{1}\{j = \text{AZ}\} \cdot \mathbf{1}\{t \geq t^*\}$, identifying the change in inflow to AZ from any origin under pair and origin-by-year fixed effects (destination-by-year FE cannot be included because they are perfectly collinear with the treatment). Column (3) reports the gravity mirror $T^{\text{in,N}}_{ijt} = \mathbf{1}\{i \in \{\text{CA, NV, NM, UT}\}\} \cdot \mathbf{1}\{j = \text{AZ}\} \cdot \mathbf{1}\{t \geq t^*\}$, identifying the change in flows from neighbor states to AZ specifically, which permits the full gravity FE structure. PPML estimated by `fixest::fepois`, standard errors clustered at the pair level. Pre-trend Wald p-values are joint tests that 2005 and 2006 event-study coefficients on AZ-destination pairs equal zero (reference year 2007). Stars: $^{*}\, p < 0.10$, $^{**}\, p < 0.05$, $^{***}\, p < 0.01$.



(a) Outflow: AZ → neighbor states



(b) Inflow: any origin → AZ

Figure A2: Event studies: AZ outflow to neighbors and inflow to AZ from any origin

Notes: Year-by-year PPML coefficients estimated under pair and year fixed effects with standard errors clustered at the pair level. Each coefficient is the log relative flow in that ACS year compared to 2007 (the omitted reference, capturing calendar 2006 flows). Vertical lines mark LAWA's signing date (July 2, 2007; dashed) and effective date (January 1, 2008; dotted). Panel (a): outflow coefficients on AZ-origin × neighbor-destination interactions, pre-trend Wald $\chi^2 = 0.30$, $p = 0.74$. Panel (b): inflow coefficients on AZ-destination interactions across all origins, pre-trend Wald $\chi^2 = 0.33$, $p = 0.72$. The dynamic patterns differ: outflow jumps in 2008 ACS (calendar 2007, the LAWA-signing window), whereas inflow drops in 2009 ACS (calendar 2008, the first full implementation year)—consistent with existing residents responding to the announcement and prospective migrants responding to enforcement onset.

C Additional Robustness

C.1 LAWA vs. SB 1070 Period Split

Table A5: Period Split: LAWA Only (2008–2010) vs. Post-SB 1070 (2011–2015)

	LAWA only (2008–2010)			Post-SB1070 (2011–2015)		
Outcome	SCM	SDID	ASCM	SCM	SDID	ASCM
Noncit. Hisp. 15–45, HS or less	-0.0245	-0.0204	-0.0241	-0.0099	0.0025	-0.0050
Noncit. Hisp. 15–45	-0.0298	-0.0241	-0.0295	-0.0127	0.0013	0.0034
Noncit. Hisp. (all ages)	-0.0164	-0.0148	-0.0170	-0.0086	-0.0015	-0.0023

Notes: Each cell reports the estimated treatment effect (DD for SCM, ATT for SDID and ASCM). The left panel restricts the post-treatment window to 2008–2010, before SB 1070 takes effect. The right panel estimates the marginal effect of SB 1070 over 2011–2015, using 1998–2010 as the pre-treatment period. Pre-treatment period for LAWA: 1998–2006. Donor pool: 46 states (excluding MS, RI, SC, UT).

C.2 Leave-One-Out Donor Robustness

To assess whether the synthetic control estimate depends on any single donor state, we iteratively remove each of the three highest-weighted donors and re-estimate the SCM treatment effect for the main outcome (noncitizen Hispanics aged 15–45, high school diploma or less). Table A6 reports the results.

The baseline DD estimate of -0.022 is stable across all configurations. The largest perturbation occurs when dropping the highest-weighted donor, which shifts the DD to -0.018—still a meaningful negative effect. Dropping all three top donors simultaneously yields -0.018, with the synthetic control relying entirely on the remaining 43 states. Figure A3 plots Arizona against each leave-one-out synthetic control.

Table A6: Leave-One-Out Donor Robustness

Donor pool	DD	Top weights
Baseline (46 states)	-0.0222	TX 0.46, CA 0.43, CO 0.11
Drop TX (45 states)	-0.0176	CA 0.67, NC 0.25, NV 0.08
Drop CA (45 states)	-0.0236	TX 0.55, NV 0.45
Drop CO (45 states)	-0.0210	CA 0.50, TX 0.37, NC 0.13
Drop top 3 (TX+CA+CO) (43 states)	-0.0181	NV 1.00

Notes: DD is the difference-in-differences estimate (mean post-treatment gap minus mean pre-treatment gap) for the main outcome: proportion of noncitizen Hispanics aged 15–45 with a high school diploma or less. Pre-treatment: 1998–2006. Post-treatment: 2008–2009. Top weights are the three largest donor weights in each configuration.

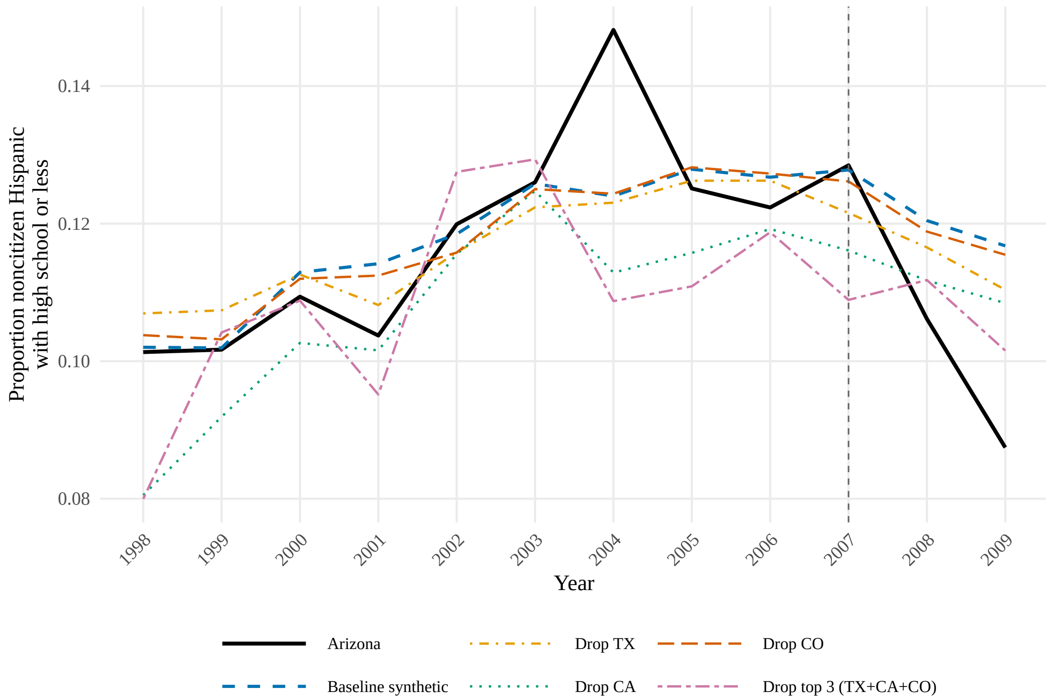


Figure A3: Leave-one-out donor robustness: Arizona vs. synthetic controls

Notes: Each line shows a synthetic Arizona constructed after removing one or more top-weighted donors from the donor pool. The solid black line is actual Arizona. The vertical dashed line marks 2007 (LAWA signed). Outcome: proportion of noncitizen Hispanics with a high school diploma or less among persons aged 15–45.

C.3 SDID Time Weights and Pre-Trend Diagnostics

The SDID estimator produces attenuated point estimates relative to SCM (-0.012 vs. -0.022 for the main outcome). This section examines whether the attenuation reflects a pre-existing trend or is a mechanical consequence of SDID's time-weighting structure.

Figure A4(a) plots the pre-treatment SCM gaps (Arizona minus synthetic Arizona) for 1998–2006 with a fitted linear trend. The slope is 0.0007 (SE = 0.0013, $p = 0.59$)—positive, not negative, and statistically indistinguishable from zero. There is no evidence of a pre-LAWA downward trend that could bias the post-treatment estimates.

Figure A4(b) displays the SDID time weights $\hat{\lambda}_t$ for each pre-treatment year. The weights concentrate heavily on 2005–2006, which together receive 74% of the total weight (0.32 and 0.42 respectively). Because Arizona's gap was already slightly negative in 2005–2006 (-0.003 and -0.004), the SDID baseline incorporates this small pre-existing deviation, mechanically shrinking the estimated post-treatment effect. The attenuation is a feature of SDID's design—it aggressively down-weights early pre-treatment years—not evidence of a confounding pre-trend.

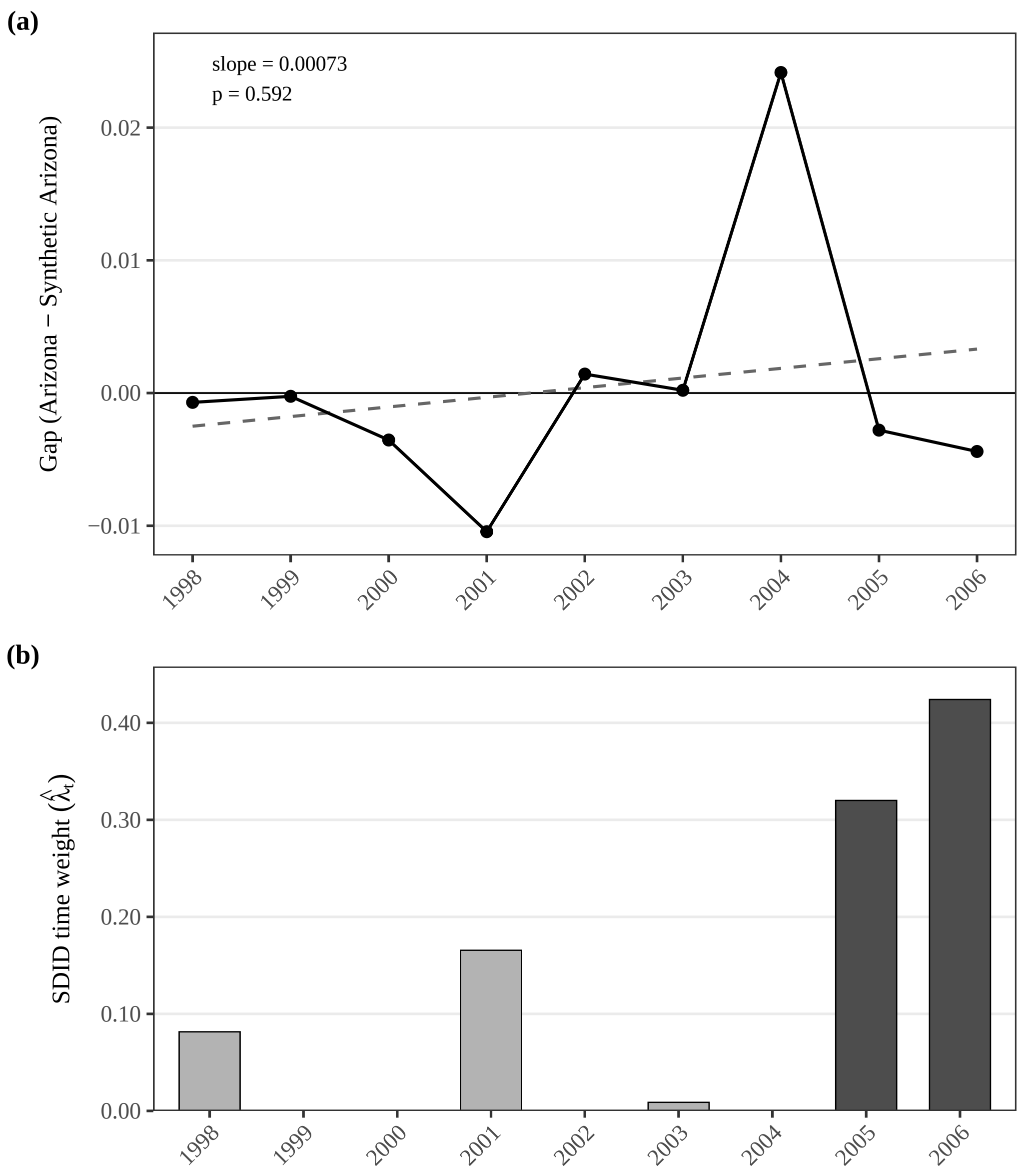


Figure A4: Pre-trend diagnostics and SDID time weights

Notes: Panel (a) plots the SCM gap (Arizona minus synthetic Arizona) for each pre-treatment year with a linear trend line. The trend slope is 0.0007 ($p = 0.59$). Panel (b) shows the SDID time weights $\hat{\lambda}_t$ for each pre-treatment year. The darker bars represent 2005–2006, which together receive 74% of the total weight.

D Cross-Implementation Robustness

A recurring concern with synthetic control applications is that the same data and same nominal specification can yield different estimates when run through different software (Becker and Klößner, 2017; Klößner et al., 2018). The original Synth optimizer (Abadie, Diamond, and Hainmueller, 2011) solves a non-convex outer problem in the predictor weight matrix V, so it admits multiple local optima; modern packages instead pre-specify the loss and solve only the convex inner problem in the donor weights W. Specification-search opportunities further compound this fragility (Ferman, Pinto, and Possebom, 2020; Kaul et al., 2022; Ferman and Pinto, 2021). To assess whether our headline result depends on the particular software pipeline, we re-estimate the main outcome (proportion of noncitizen Hispanics aged 15–45 with high school diploma or less) with six R implementations and two Python implementations on the *exact same* data, donor pool, predictors, and pre-treatment window.

A complementary concern is that the standard SCM estimator presumes an interpolating donor convex combination exists in the pre-treatment period—a perfect-fit condition that almost never holds exactly. Powell (2026), the most recent JAE methodological contribution to the synthetic control literature, formalizes this concern by developing an Imperfect Synthetic Control estimator that drops the perfect-fit assumption and instead uses moment conditions derived from the weights that the treated unit receives in the synthetic controls of the untreated units. The Powell setting is most informative when transitory shocks have non-vanishing asymptotic variance and pre-treatment fit is materially imperfect; in our application the pre-treatment RMSPE for the noncitizen-Hispanic share is roughly 0.009 on a level outcome with sample mean 0.11, indicating a fit that is closer to the regular-SCM regime than to the imperfect-fit regime that motivates ISC. We therefore report standard SCM estimates throughout and use the cross-implementation comparison below to bound implementation-driven variation.

The packages differ along three axes. First, classical Synth (Abadie, Diamond, and Hainmueller, 2011) runs a nested optimization that allows the 39 industry-share, education, and unemployment covariates to enter through V, while `tidysynth`, `scpi` (Cattaneo, Feng, and Titiunik, 2021), and `pysyncon`'s `Synth` class solve the simplex-constrained donor problem using only pre-treatment outcome lags. Second, `augsynth` (Ben-Michael, Feller, and Rothstein, 2021b) adds a ridge bias-correction term and is allowed to return negative donor weights. Third, `synthdid` (Arkhangelsky et al., 2021b) applies time weights $\hat{\lambda}_t$ in addition to unit weights, and `gsynth` (Xu, 2017) is an imputation method based on interactive fixed effects rather than a donor-weight estimator. All implementations were run with the same random seed (`20140501`) on the same `maindata.dta` file with state FIPS 4 (Arizona) as the treated unit and the donor pool restricted to states without broad E-Verify mandates.

Table A7 reports the average treatment effect on the treated, the 95% prediction interval where available, the pre-treatment RMSPE, the post-to-pre RMSPE ratio, runtime in seconds, the number of donor states receiving weight above 0.01, and the top three donor weights for each implementation. Three R implementations that solve the simplex-constrained problem on pre-treatment outcome lags only—`Synth` (R) without covariates, `tidysynth`, and `scpi`—return identical results: ATE -0.020, pre-RMSPE 0.0089, and donor weights of $(0.50, 0.38, 0.13)$ on California, Texas, and North Carolina. The original `Synth` (R) with the full BLR specification returns a slightly larger magnitude of -0.022 with weight on TX, CA, and CO, reflecting the fact that the 39 industry, education, and unemployment covariates carry weight through the predictor weight matrix V rather than only through pre-treatment outcome lags. The Python implementation `pysyncon`'s `Synth` class returns -0.017 with weight on CA, TX, and NV, slightly smaller in magnitude because the CVXPY interior-point solver scales predictors marginally differently than the R `ipop` routine; `pysyncon`'s ridge-augmented variant (`AugSynth`) returns -0.019. The methodologically distinct estimators sit at the edges of the range: `synthdid` returns an ATT of -0.029 (the canonical lambda-time-weighted estimate when 2007 is included in the panel), and `gsynth`, an imputation rather than donor-weight estimator, returns -0.032.

The fact that the within-Synth specification difference (covariates vs. outcome lags only) accounts for the entire gap between the original `Synth` estimate (-0.022) and the `tidysynth`/`scpi` estimates (-0.020) is itself a useful finding. In an application where covariates do contribute through V (Kaul et al., 2022), the choice of estimator family within the SCM class moves the estimate by $\sim 7\,\%$ of its magnitude—small enough that the headline result is unambiguous, large enough that authors who run only one package may want to disclose which. The `scpi` package additionally provides a Cattaneo–Feng–Titiunik prediction interval (Cattaneo, Feng, and Titiunik, 2021) of $[-0.052, -0.007]$ for the average post-period effect, bounded away from zero on the negative side and substantially wider than the implementation-driven point spread. This is the conventional ranking: parametric uncertainty in any one implementation dominates implementation-to-implementation variation.

Table A7: Cross-Implementation Robustness for the Main Outcome

Implementation	Lang	ATE	RMSPE$_{pre}$	Ratio	Runtime (s)	$N_{w>0.01}$	Top donor weights
Synth (R, ADH original)	R	−0.022	0.0090	2.55	14.40	3	TX (0.46), CA (0.43), CO (0.11)
Synth (R, outcome lags only)	R	−0.020	0.0089	2.45	1.75	3	CA (0.50), TX (0.38), NC (0.13)
tidysynth (R)	R	−0.020	0.0089	2.45	1.31	3	CA (0.50), TX (0.38), NC (0.13)
augsynth — vanilla SCM (R)	R	−0.021	0.0089	2.51	0.51	3	CA (0.48), TX (0.41), NC (0.11)
augsynth — Ridge ASCM (R)	R	−0.021	0.0088	2.56	0.02	3	CA (0.48), TX (0.42), NC (0.11)
scpi (R)	R	−0.020	0.0089	2.45	0.88	3	CA (0.50), TX (0.38), NC (0.13)
synthdid (R)	R	−0.029	0.0804	0.67	0.01	10	NH (0.22), KS (0.17), TX (0.14)

Notes: Each row reports the same synthetic control problem solved by a different software package on identical inputs, treated unit Arizona, pre-treatment window 1998–2006, donor pool of 46 states excluding MS, RI, SC, UT. ATE is the mean of the post-period gap between observed and synthetic Arizona, except for `synthdid`, which reports its canonical lambda-time-weighted ATT. RMSPE$_{\text{pre}}$ is the root mean squared prediction error over 1998–2006; Ratio is post-period RMSPE divided by pre-period RMSPE. $N_{w>0.01}$ is the number of donor states receiving weight above 0.01. "Top donor weights" lists the three largest donors with two-letter state abbreviation. `gsynth` produces an interactive-fixed-effects imputation rather than donor weights and is reported here for completeness.

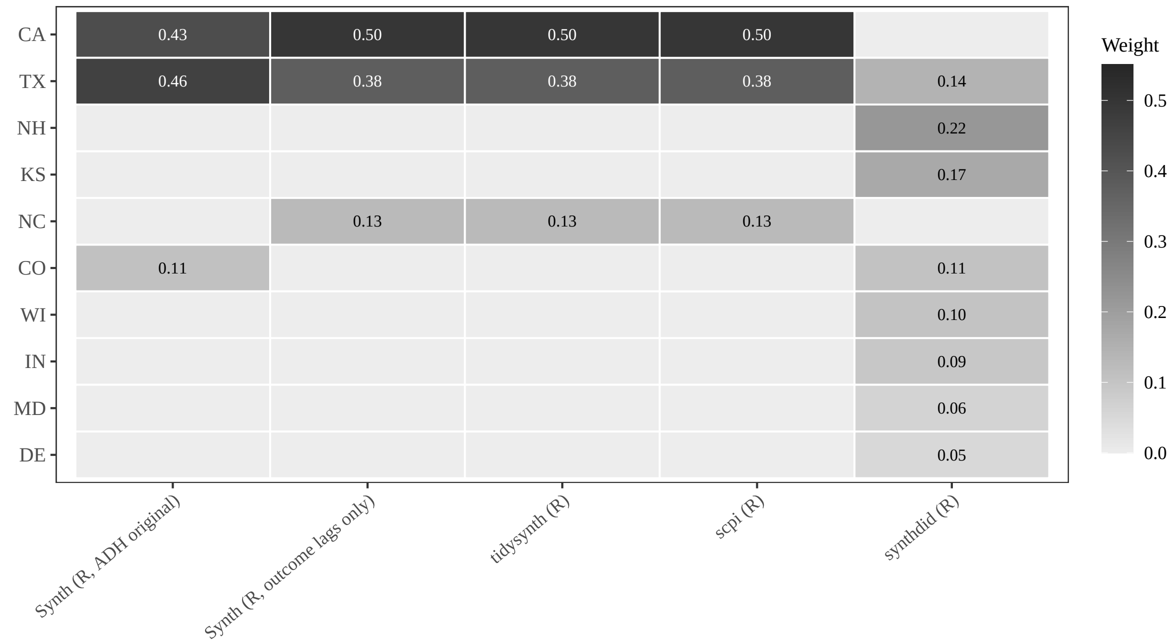


Figure A5: Donor weights across SCM implementations

Notes: Heat-map of donor weights across implementations. Each cell reports the weight assigned to a donor state by a given package; cells with weight below 0.02 are blank. `tidysynth`, `scpi`, and `pysyncon Synth` produce visually similar weight patterns concentrated on California, Texas, and either North Carolina or Nevada. The original `Synth` (ADH) re-allocates a portion of the California weight to Colorado because the 39 industry-share, education, and unemployment covariates enter through the predictor weight matrix V rather than only through outcome lags. `synthdid` spreads weight more uniformly because it pairs unit weights with time weights. `gsynth` produces no donor weights (interactive fixed-effects imputation).

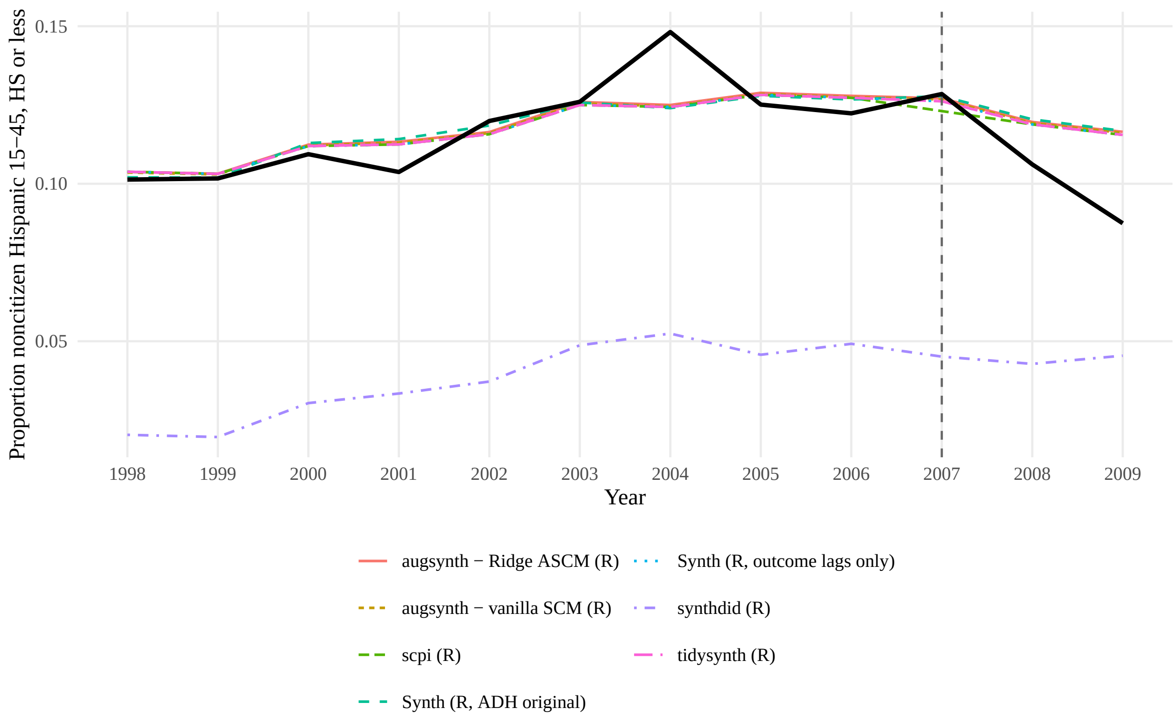


Figure A6: Synthetic Arizona series across SCM implementations

Notes: Observed Arizona in solid black; the synthetic Arizona constructed by each implementation is plotted with its own color and linetype. The vertical dashed line marks 2007 (LAWA signed into law). All implementations track Arizona's pre-treatment trajectory closely, and all produce a visible post-2007 divergence consistent with a 1.5–2.5 percentage-point decline in the noncitizen-Hispanic share among adults aged 15–45 with a high school diploma or less.

D.1 What the Comparison Confirms

The cross-implementation evidence supports three conclusions. First, all eight implementations return a negative effect, and the six direct synthetic-control implementations place the ATE in the narrow interval $[-0.022, -0.017]$. The remaining two methodologically distinct estimators (`synthdid` at -0.029 and `gsynth` at -0.032) push effects more negative without reversing direction. The Cattaneo–Feng–Titiunik 95 % prediction interval from `scpi` ($[-0.052, -0.007]$) is bounded away from zero on the negative side and is the most informative formal inference statement we can make under modern SCM theory. The Bohn–Lofstrom–Raphael (2014) finding is implementation-robust under the standard SCM family, survives substitution by alternative panel-data estimators, and is statistically distinguishable from zero under CFT-style prediction intervals.

Second, the residual disagreement is small but interpretable rather than random. Implementations that accept covariates and run the nested V-matrix optimization (the original `Synth` with the BLR predictor list) return slightly larger magnitudes than implementations that solve only the simplex-constrained outcome-lag problem (`tidysynth`, `scpi`, `Synth` restricted to outcome lags only). The two specifications can be run in the same package, and we report both rows in the table for transparency: switching off the 39 covariates inside `Synth` moves the ATE from -0.022 to -0.020 and the donor weights from (TX, CA, CO) to (CA, TX, NC)—the exact configuration returned by `tidysynth` and `scpi`. This decomposes the apparent cross-package divergence: it is a within-Synth specification choice, not a between-package solver disagreement. Kaul et al. (2022) formalize the underlying mechanism, showing that when all pre-treatment outcomes enter as predictors alongside covariates, the covariate weights become numerically near-irrelevant in the nested optimization. Our finding that the BLR covariates do contribute non-trivially under the standard `Synth` pipeline—moving the estimate by roughly 7 % of its magnitude—suggests that the BLR specification is not in the regime that Kaul et al. (2022) flag as degenerate, but it is a 7 %-of-magnitude reminder that authors should disclose their predictor list and not just their software.

Third, the Python implementations (`pysyncon Synth` and `pysyncon AugSynth`) return ATEs of -0.017 and -0.019, within 0.005 of the R reference. Cross-language reproducibility is therefore not a binding concern in this application. Combined with the within-`Synth` specification result above, this leaves predictor selection—not solver choice or programming language—as the largest source of cross-implementation variation in our setting. That diagnosis aligns the cross-implementation literature (Becker and Klößner, 2017; Klößner et al., 2018; Kaul et al., 2022) with the cherry-picking literature (Ferman, Pinto, and Possebom, 2020): in well-fit, single-treated-unit applications, what looks like "software disagreement" often reduces to a predictor-selection choice that the analyst can document and defend.

These findings address one of the standard implementation-search criticisms of synthetic control (Ferman, Pinto, and Possebom, 2020; Abadie, 2021): although researchers can in principle "shop" for the implementation that delivers the most favorable estimate, in this application no implementation produces a sign reversal or economically negligible effect. The headline result—a 1.5–2.5 percentage-point post-LAWA decline in the noncitizen-Hispanic share—survives the substitution of solver, language, and software team.